

PROVISIONAL PATENT APPLICATION

REMOVABLE PASSIVE RESERVOIR INSERT FOR A PLANTER

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TITLE OF THE INVENTION

Removable Passive Reservoir Insert for a Planter

FIELD OF THE INVENTION

The present disclosure relates generally to horticultural containers and, more particularly, to a removable support insert configured to create a water-storage region beneath growing medium within an existing planter.

BACKGROUND OF THE INVENTION

Conventional planters rely on periodic watering and drainage through the bottom of the container. Depending on the planter geometry, growing medium, and watering frequency, water may drain too quickly, remain unevenly distributed, or collect around the root zone. Extended root contact with standing water can contribute to poor plant health, while rapid drainage can require frequent watering.

Self-watering planters and reservoir systems are available, but many require a dedicated planter body, multiple fitted components, powered equipment, or replacement of a container already owned by the user. A need therefore exists for a simple, removable insert capable of creating a passive reservoir region within a conventional planter while supporting growing medium above the stored water.

SUMMARY OF THE INVENTION

The invention provides a removable planter insert having an outer support rim and an open interior support structure. When positioned above the bottom of a planter, the insert supports growing medium and a plant root zone while defining a lower water-storage region. Openings through the support structure permit drainage and passive movement of moisture between the growing medium and the lower region.

In one embodiment, the open interior support structure includes a plurality of concentric rings connected to the outer rim by intersecting braces. The structure distributes the supported load while limiting material use and preserving open flow paths. The insert may be manufactured as a unitary component by additive manufacturing and may be removed for cleaning, reuse, replacement, or transfer to another compatible planter.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a top engineering view of a Revision 5 planter insert having an overall diameter of approximately 4.25 inches.

FIG. 2 is a side engineering view of the Revision 5 planter insert.

FIG. 3 is a multi-view engineering drawing of an earlier Iteration 1.4 planter insert.

FIG. 4 is a rendered perspective view of the planter insert.

FIG. 5 is an engineering simulation image associated with an example design iteration.

DETAILED DESCRIPTION OF THE INVENTION

A removable planter insert includes a generally circular outer rim surrounding an open, load-supporting interior. In the illustrated embodiment, the interior includes spaced concentric rings joined by first and second braces that intersect near the center of the insert. The braces extend across the ring structure and connect the interior structure to the outer rim. Open spaces remain between adjacent rings and braces.

The insert is sized to be received within the lower portion of a conventional planter. Revision 5 has an overall diameter of approximately 4.25 inches and was developed for use within an approximately six-inch planter. These dimensions describe one embodiment only. The insert may be scaled or parametrically modified to accommodate planters having different diameters, tapers, bottom contours, wall thicknesses, and reservoir-depth requirements.

During installation, the insert is positioned above the bottom surface of the planter. The outer rim and interior structure support growing medium placed above the insert. The spacing between the insert and the planter bottom defines a reservoir region. Water introduced from above passes through the growing medium and the open insert into the reservoir region. Moisture may subsequently move upward through contact with the growing medium, a capillary element, or limited root growth through the openings.

The insert separates most of the supported root zone from direct and continuous contact with water stored at the bottom of the planter. At the same time, the openings allow drainage, airflow, and moisture interaction. Because the insert is not permanently integrated into the planter body, it can be removed without destroying the planter and can be cleaned or replaced independently.

The insert may be formed from a polymer suitable for additive manufacturing. Other embodiments may be molded, machined, cast, assembled from multiple pieces, or formed from corrosion-resistant metal, composite, ceramic, or other water-compatible material. The ring and brace dimensions may be selected according to material properties, anticipated growing-medium load, print orientation, manufacturing process, and desired open area.

The illustrated concentric-ring arrangement is one example. Other embodiments may use lattice, perforated plate, radial ribs, polygonal cells, curved ribs, or a combination thereof. The outer support portion may be circular, oval, polygonal, segmented, flexible, or adjustable. Feet, spacers, or a peripheral ledge may establish the height of the reservoir region.

Optional embodiments may include one or more capillary wicks, a fill tube, an overflow passage, a drain opening, a removable filter, or a visible water-level indicator. A further embodiment may include a water-level sensor, soil-moisture sensor, temperature sensor, microcontroller, indicator, display, or communications module. These optional components are not required for operation of the passive mechanical embodiment.

A static stress study was performed on an example CAD model. The software displayed a safety factor of 15 for the defined model and load case, indicating that the evaluated geometry may permit material reduction. The displayed value applies only to that simulation and is not a universal load rating for every material, manufacturing condition, planter, or use case.

EXAMPLE METHOD OF USE

A user selects an insert sized for a planter, places the insert above the planter bottom, adds growing medium above the insert, positions a plant in the growing medium, and introduces water. At least a portion of the water passes through the growing medium and insert into the lower reservoir region. The insert supports the growing medium above the reservoir while permitting passive moisture transfer.

CLAIMS

1. A removable insert for a planter, comprising: an outer support portion configured to be received within the planter; and an open interior support structure connected to the outer support portion and configured to support growing medium above a bottom region of the planter, thereby defining a water-storage region below the growing medium while permitting water movement through the open interior support structure.
2. The removable insert of claim 1, wherein the open interior support structure comprises a plurality of spaced concentric rings.
3. The removable insert of claim 2, further comprising intersecting braces connecting the plurality of spaced concentric rings to the outer support portion.
4. The removable insert of claim 1, wherein the outer support portion and the open interior support structure form a unitary additively manufactured component.
5. The removable insert of claim 1, wherein the insert is removable from the planter for cleaning, reuse, replacement, or transfer to another planter.
6. The removable insert of claim 1, further comprising a capillary element configured to transfer water from the water-storage region toward the growing medium.
7. The removable insert of claim 1, further comprising at least one of a fill passage, an overflow passage, a drain passage, or a water-level indicator.
8. The removable insert of claim 1, further comprising a water-level sensor or a soil-moisture sensor.
9. A planter system comprising: a planter having a bottom region; the removable insert of claim 1 positioned above the bottom region; growing medium supported above the removable insert; and a water-storage region located between the removable insert and the bottom region.
10. A method of forming a passive reservoir in a planter, comprising: positioning a removable open support insert above the bottom of the planter; supporting growing medium above the insert; and receiving water through the growing medium and the insert within a reservoir region located below the insert.

ABSTRACT

A removable passive reservoir insert is configured for placement within an existing planter. The insert includes an outer support portion and an open interior support structure that supports growing medium and a plant root zone above a lower water-storage region. In one embodiment, the interior structure includes spaced concentric rings connected by intersecting braces. Openings permit drainage and passive moisture movement while the insert separates most of the supported root zone from continuous contact with stored water. The insert may be manufactured as a unitary additively manufactured component and removed for cleaning, reuse, resizing, or replacement. Optional embodiments may include capillary elements, fill or overflow passages, water-level indicators, or embedded sensing components.

DRAWINGS:

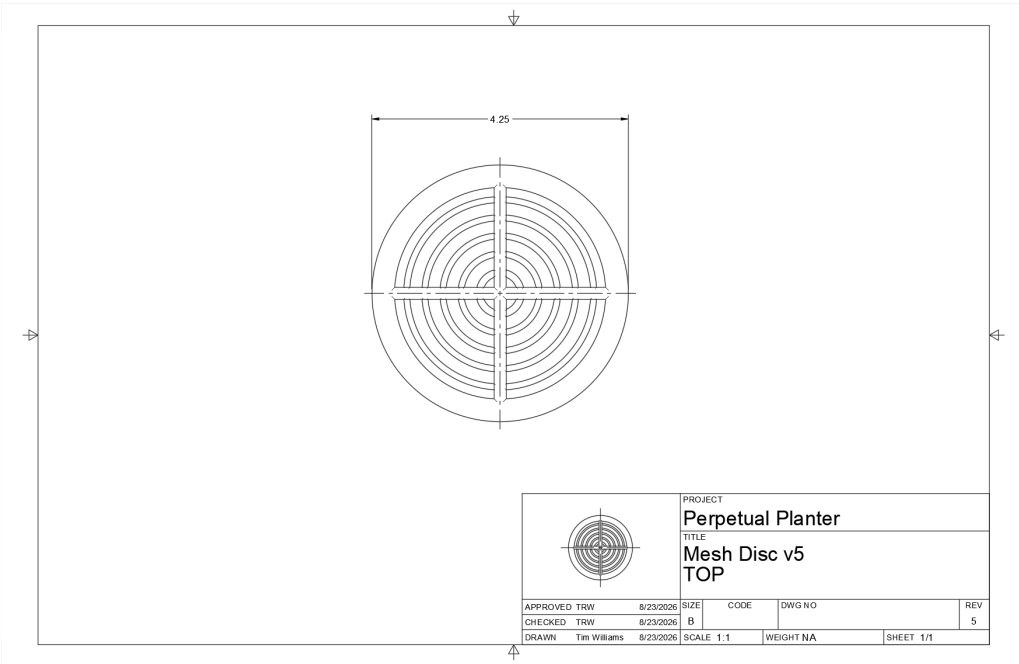


FIG. 1 — Revision 5 Top View

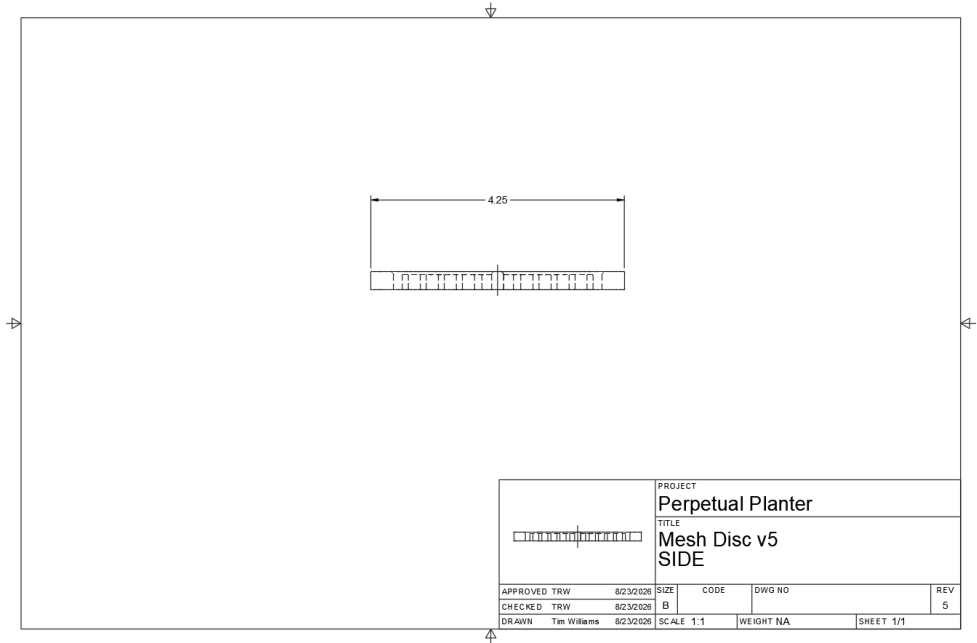


FIG. 2 — Revision 5 Side View

DRAWINGS:

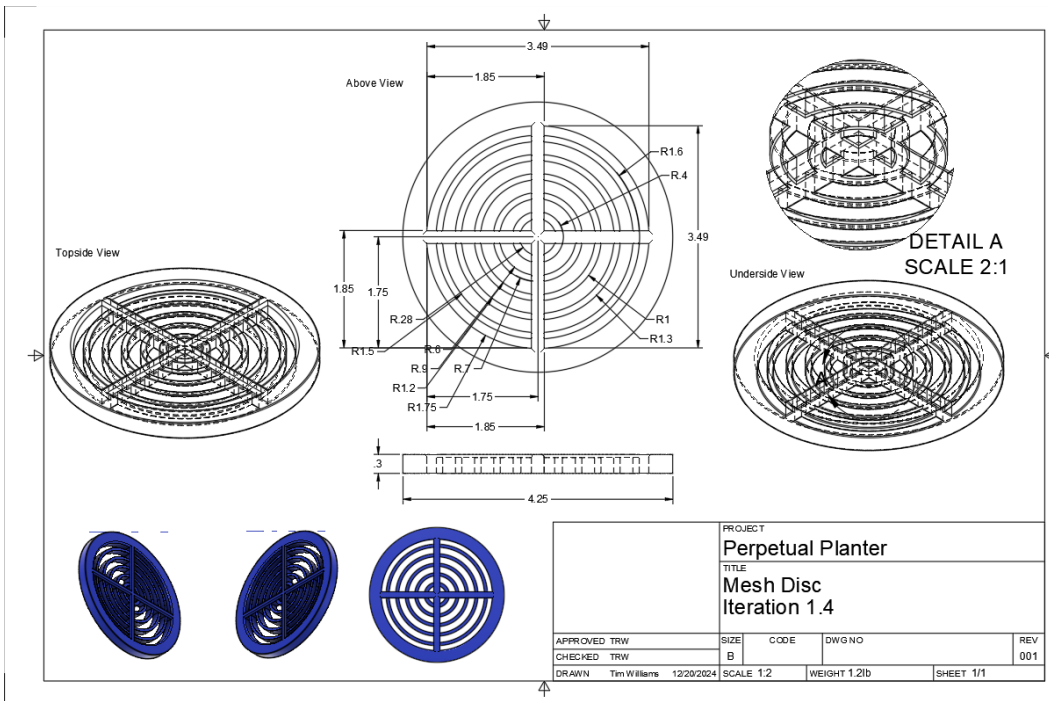


FIG. 3 — Iteration 1.4 Multi-View Drawing



FIG. 4 — Perspective Rendering

DRAWINGS:

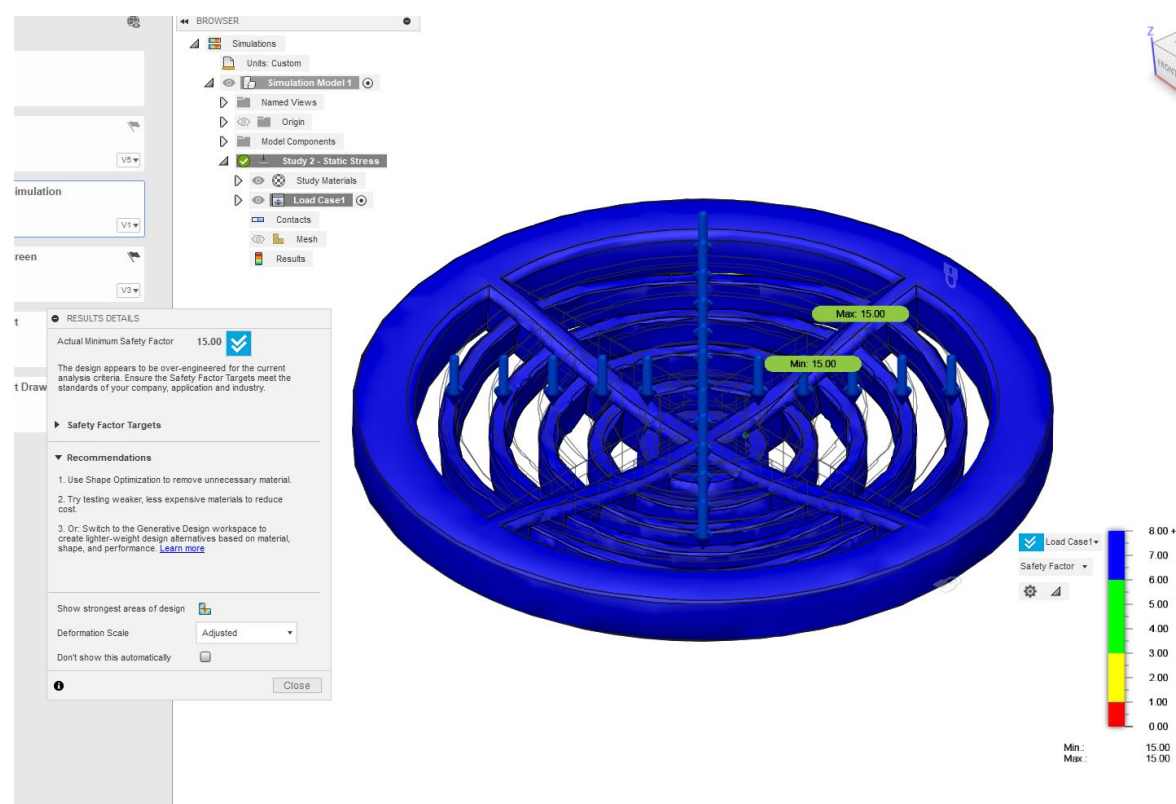


FIG. 5 — Example Static Stress Study